

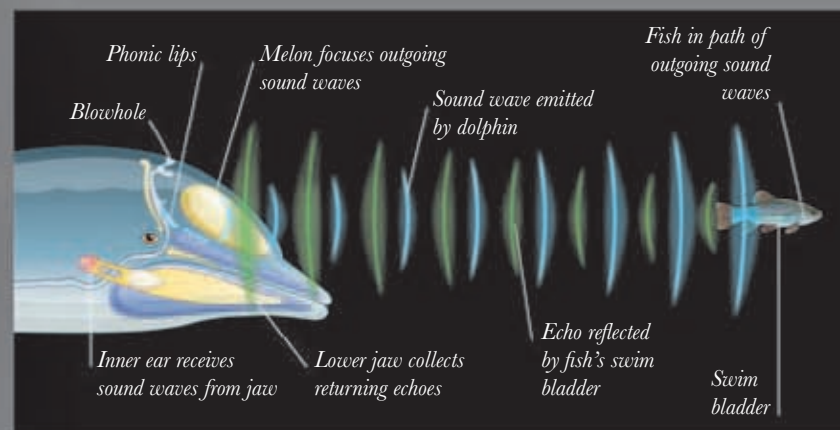
Cave chorus

In Southeast Asia, cave swiftlets often roost and nest deep in limestone caves. They have good eyesight, but they use echolocation to find their way once they are underground. They make clicking sounds normally about 10 times a second, but in narrow passageways, the rate speeds up, giving the swiftlets a better picture of the route ahead.



Sound in the sea

Sound travels much better in water than it does in air, and it even penetrates the seabed. Ocean dolphins use it to find the echoes of flatfish hiding in the sand. They communicate with noisy whistles and squeaks, but their echolocation clicks can have a frequency of more than 150,000 Hz, which is far beyond the range of human hearing.



Dolphin clicks

Dolphins make click sounds with phonic lips—valves inside their heads, beneath their blowholes. They focus sound waves with the melon—a fatty pad that gives them a bulging forehead. Water lets sound pass, but a fish's gas-filled swim bladder reflects the sound, producing an echo.

Echoes in the deep

The sperm whale is the largest toothed whale and also the largest predator that hunts individual prey. It dives far into the sea's depths, where it uses sound to track down octopuses and squid. It produces the loudest and deepest echolocation clicks made by any animal, focusing them with spermaceti—a waxy substance that also adjusts the whale's buoyancy when diving.



Whale's enormous, square-fronted head is dominated by its spermaceti organ